

1 Introduction to Categories

1.1 Motivation

Many areas of mathematics have the same basic pattern:

- Objects (such as groups, sets, vector spaces, topological spaces).
- Structure-preserving maps between those objects.

Examples:

Objects	Structure-Preserving Maps
Sets	Functions
Groups	Group Homomorphisms
Vector Spaces	Linear Transformations
Topological Spaces	Continuous Functions

Category theory studies mathematical structures through the relationships (maps) between them.

1.2 Definition of a Category

A **category** $\mathcal{C}$ consists of:

1. A collection of objects.
2. For every pair of objects A, B , a collection of morphisms (or arrows)

$$\text{Hom}_{\mathcal{C}}(A, B),$$

whose elements are written

$$f : A \rightarrow B.$$

3. A composition operation

$$\text{Hom}(A, B) \times \text{Hom}(B, C) \longrightarrow \text{Hom}(A, C),$$

given by

$$(f, g) \mapsto g \circ f.$$

4. For every object A , an identity morphism

$$1_A : A \rightarrow A.$$

These data must satisfy the following axioms.

1.3 Category Axioms

1.3.1 Associativity

For morphisms

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D,$$

we require

$$(h \circ g) \circ f = h \circ (g \circ f).$$

1.3.2 Identity

For every morphism

$$f : A \rightarrow B,$$

we have

$$f \circ 1_A = f,$$

and

$$1_B \circ f = f.$$

1.4 Example: The Category **Set**

The category **Set** has:

- Objects: Sets.
- Morphisms: Functions between sets.

Composition is ordinary function composition, and identity morphisms are identity functions.

Therefore **Set** is a category.

1.5 Example: The Category **Grp**

The category **Grp** has:

- Objects: Groups.
- Morphisms: Group homomorphisms.

For example,

$$f : \mathbb{Z} \rightarrow \mathbb{Z}_6, \quad f(n) = n \pmod{6},$$

is a group homomorphism.

Composition is composition of homomorphisms, and identity morphisms are identity homomorphisms.

Therefore **Grp** is a category.

1.6 Example: The Category $\mathbf{Vect}_F$

The category $\mathbf{Vect}_F$ has:

- Objects: Vector spaces over a field F .
- Morphisms: Linear transformations.

Example:

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad T(x, y) = (x + y, y).$$

Since T is linear, it is a morphism in $\mathbf{Vect}_{\mathbb{R}}$.

1.7 Terminology

Classical Mathematics	Category Theory
Set	Object
Function	Morphism
Homomorphism	Morphism
Linear Transformation	Morphism
Continuous Function	Morphism

In category theory, the term *morphism* refers to the appropriate structure-preserving map.

1.8 Why Morphisms Matter

A central philosophy of category theory is:

To understand an object, study how it relates to other objects through morphisms.

For example, instead of studying a group G only through its internal structure, one may study collections of morphisms

$$\mathrm{Hom}(G, H)$$

for various groups H .

1.9 Diagrams

Categories are often represented using diagrams:

$$A \xrightarrow{f} B \xrightarrow{g} C.$$

Composition produces the morphism

$$A \xrightarrow{g \circ f} C.$$

Reasoning with such diagrams is fundamental in category theory.

1.10 A Small Category

Consider a category with objects

$$\{A, B, C\}$$

and morphisms

$$1_A, 1_B, 1_C, f : A \rightarrow B, g : B \rightarrow C.$$

Closure under composition requires the morphism

$$g \circ f : A \rightarrow C.$$

This forms a category.

1.11 Key Idea

A category consists of:

- Objects,
- Morphisms between objects,

- Composition of morphisms,
- Identity morphisms,

satisfying associativity and identity laws.

The primary focus is not the objects themselves, but the network of morphisms connecting them.

2 Generators in a Category

2.1 Definition

Let $\mathcal{C}$ be a category. An object $G \in \mathcal{C}$ is called a *generator* if for every pair of distinct morphisms

$$f, g : X \rightarrow Y,$$

there exists a morphism

$$u : G \rightarrow X$$

such that

$$f \circ u \neq g \circ u.$$

Equivalently, whenever

$$f \circ u = g \circ u$$

for all morphisms $u : G \rightarrow X$, it follows that

$$f = g.$$

Thus, maps from G are sufficient to distinguish morphisms in the category.

2.2 Intuition

A generator is an object that acts as a *probe*. If two morphisms differ, then some map from the generator can detect the difference.

In many categories, morphisms from a generator correspond to the “elements” or “points” of an object.

2.3 Examples

1. The Category of Sets

In the category **Set**, any singleton set

$$\{*\}$$

is a generator.

A function

$$u : \{*\} \rightarrow X$$

is completely determined by the image of $*$, so there is a natural bijection

$$\mathrm{Hom}_{\mathbf{Set}}(\{*\}, X) \cong X.$$

If

$$f, g : X \rightarrow Y$$

are distinct, then there exists $x \in X$ such that

$$f(x) \neq g(x).$$

Define

$$u(*) = x.$$

Then

$$(f \circ u)(*) = f(x) \quad \text{and} \quad (g \circ u)(*) = g(x),$$

so

$$f \circ u \neq g \circ u.$$

Hence $\{*\}$ is a generator of **Set**.

2. The Category of Groups

In the category **Grp**, the group

$$\mathbb{Z}$$

is a generator.

If

$$f, g : G \rightarrow H$$

are distinct group homomorphisms, then there exists $a \in G$ such that

$$f(a) \neq g(a).$$

Define

$$u : \mathbb{Z} \rightarrow G, \quad u(n) = a^n.$$

Then

$$(f \circ u)(1) = f(a) \quad \text{and} \quad (g \circ u)(1) = g(a),$$

so

$$f \circ u \neq g \circ u.$$

Therefore $\mathbb{Z}$ is a generator of **Grp**.

3. The Category of Vector Spaces

Let $\mathbf{Vect}_F$ denote the category of vector spaces over a field F .

The one-dimensional vector space

$$F$$

is a generator.

For every vector space V ,

$$\mathrm{Hom}_{\mathbf{Vect}_F}(F, V) \cong V.$$

Indeed, a linear map

$$u : F \rightarrow V$$

is determined by the vector

$$u(1) \in V.$$

If

$$T, S : V \rightarrow W$$

are distinct linear maps, there exists $v \in V$ such that

$$T(v) \neq S(v).$$

Define

$$u(a) = av.$$

Then

$$(T \circ u)(1) = T(v) \quad \text{and} \quad (S \circ u)(1) = S(v),$$

so

$$T \circ u \neq S \circ u.$$

Hence F is a generator of $\mathbf{Vect}_F$.

4. The Category of R -Modules

Let $R\text{-}\mathbf{Mod}$ denote the category of left R -modules.

The module

$$R$$

(viewed as a left module over itself) is a generator.

For every R -module M ,

$$\mathrm{Hom}_R(R, M) \cong M.$$

Indeed, every module homomorphism

$$u : R \rightarrow M$$

is completely determined by

$$u(1) \in M,$$

since

$$u(r) = r u(1).$$

If

$$f, g : M \rightarrow N$$

are distinct module homomorphisms, then there exists $m \in M$ such that

$$f(m) \neq g(m).$$

Define

$$u : R \rightarrow M, \quad u(r) = rm.$$

Then

$$(f \circ u)(1) = f(m) \quad \text{and} \quad (g \circ u)(1) = g(m),$$

so

$$f \circ u \neq g \circ u.$$

Therefore R is a generator of $R\text{-Mod}$.

5. The Category of Topological Spaces

In the category **Top**, the one-point space

$$\{*\}$$

is a generator.

A continuous map

$$u : \{*\} \rightarrow X$$

corresponds exactly to a point of X .

Thus,

$$\mathrm{Hom}_{\mathbf{Top}}(\{*\}, X) \cong X.$$

If

$$f, g : X \rightarrow Y$$

are distinct continuous maps, there exists $x \in X$ such that

$$f(x) \neq g(x).$$

Define

$$u(*) = x.$$

Then

$$f \circ u \neq g \circ u.$$

Hence $\{*\}$ is a generator of $\mathbf{Top}$.

2.4 Summary Table

Category	Generator	Maps from Generator Represent
Set	$\{*\}$	Elements
Grp	$\mathbb{Z}$	Group elements
Vect_F	F	Vectors
R-Mod	R	Module elements
Top	$\{*\}$	Points

2.5 Key Insight

A generator allows us to study objects through morphisms from a single distinguished object. In many categories, morphisms from the generator correspond precisely to the “elements” of an object.

Thus generators convert internal structure into external morphisms, a central theme of category theory.